

OXFORD

INTERNATIONAL
AQA EXAMINATIONS

INTERNATIONAL AS CHEMISTRY

(9620) CH01

Report on the examination

January 2024

REPORT ON EXAMINATION: INTERNATIONAL AS CHEMISTRY (9620) CH01 JANUARY 2024

All questions were answered by some students: the paper discriminated well. Knowledge of practical techniques was not often well known. The calculations were generally answered better than answers requiring explanations.

QUESTION 01

In part 1 many students stated metallic bonding without describing the attraction between the positive metal ions and the sea of delocalised electrons. Part 2 was generally well done; a few students did not give an equation starting from the elements. 75% of students scored both marks in part 3; the most common incorrect answer had one lone pair and a pyramidal shape. Most students scored one mark in part 4 but many lost the second mark by not being specific e.g. saying that the electrons came from one atom only or from chlorine rather than from the chloride ion. 65% of students scored the mark in part 5.

QUESTION 02

Many students knew the reagent and the observations this gave in part 1; incorrect reagents given included HCl, NH_3 and $\text{Ba}(\text{OH})_2$. In part 2, 67% of students gave a correct equation; common errors included a bromine atom gaining one electron $\text{Br} + \text{e}^- \rightarrow \text{Br}^-$ or an unbalanced equation. In part 3, 43% of students gave the correct half-equation; common errors included electrons on the wrong side of the half-equation, giving the wrong number of electrons and omitting the charge on the sulfate ion. Students lost marks in part 4 by not balancing equations, omitting some species or giving an overall equation with electrons in it; 43% of students gave a correct equation. Just over half the students gave the correct answer to part 5; the most common incorrect answer was sulfur.

The equation in part 6 was better known than the observation; common mistakes for the observation were purple gas or solution and brown solid. Some students gave unbalanced equations or gave incorrect formulae for potassium chloride/iodide (KCl_2 and KI_2) or iodine/chlorine (I and Cl).

QUESTION 03

In part 1 students often missed out one or more of the essential words e.g. mass of an atom (without average or mean) or divided by $1/12$ of a C-12 atom without mentioning the mass of C-12 atom; 29% of students scored full marks. Part 2 was well done. Many students made good progress with the calculation in part 3, however, many students did not notice that the mass of one mole of ions was required, and many gave this answer as the mass of one ion (either in grams or kilograms) and some students failed to give the mass number as an integer. Some students used a distance of 1500 cm instead of 1.5 m when calculating speed and others failed to take a square root or missed out the 2 on re-arrangement of the equation. 18% of students scored all the marks in part 3.

In part 4, 48% of students gave a correct answer; common errors included stating that the number of neutrons was the same or that the number of protons was the same without any reference to electrons.

QUESTION 04

56% of students gave a correct formula in part 1; SiC_4 , SiCO_3 were common incorrect answers. In part 2, 18% of students scored both marks. Many students stated that the molecule was tetrahedral or symmetrical but could not explain that the dipoles would cancel; some students stated that there was no lone pair so the molecule couldn't be polar. Part 3 was well done with 78% of students giving the correct answer. In part 4, 12% of students scored all the marks. Most students stated that the intermolecular

force was a hydrogen-bond, but few students gained full marks for the diagram; the hydrogen bond was often drawn to the oxygen rather than to a lone pair and few students drew a straight line for the $\text{O} \cdots \text{H}-\text{O}$. Other common errors included missing the partial charges or omitting lone pairs on the oxygen atoms. A few students drew water molecules with two oxygens and one hydrogen.

QUESTION 05

In part 1 most students continued with their working to the end of the calculation and gained some credit; 42% of students gained all the marks. Several students used 25.0 cm^3 or 2.3 cm^3 ($27.3 - 25.0$) as the volume of acid in their calculation; other common errors included not dividing the moles of acid by 2, not multiplying $\times 10$ and not subtracting 34. Part 2 was not well answered; a common incorrect answer was to zero the balance, take several readings and take an average. Some students got close to the correct answer stating the student should weigh after adding the metal hydroxide but not stating that the weighing should be done again after transferring the solid to the beaker. Common incorrect answers to part 3 included burette and pipette. 14% of students gave a correct answer to part 4; some students stated that water should be added dropwise, or water added to the mark but often failed to connect the two points. Many students answered part 5 well; some students stated which titres to use but did not explain why they were chosen. In part 6, 18% of students gave good answers to both parts of the question; common incorrect answers included stating that water does not contain any contaminants or water would not affect the concentration of the reagents.

QUESTION 06

Answers to part 1 were generally good. In part 2, 52% of students gained full marks; common errors included using the wrong A_r , incorrect conversion of temperature and/or pressure, and incorrect conversion into cm^3 .

Parts 3 and 4 were generally well answered. In part 4 common errors included the use of H_2SO_4 instead of potassium sulfate, giving incorrect formulae e.g. KSO_4 and $\text{K}(\text{OH})_2$, or giving unbalanced equations. Part 5 differentiated well; many students lost the last mark by giving the correct numerical number for their enthalpy change but did not give a negative value. Other common errors included using of 2.53 g for the mass of solution to be heated and using $5.4 + 273 = 278.4$ for the temperature rise.

QUESTION 07

Part 1 was correctly answered by 44% of students; chlorine was a common incorrect answer. In part 2 several students lost the mark by not giving all the details required e.g. missing atom, electron pair or covalent bond.

Answers to part 3 were generally good. In Part 3 most students wrote that proton number or nuclear charge increased; those who did not gain the second mark often wrote about increasing electron number but did not state that the electrons were in the same shell. Most students gave a correct answer to part 4; common incorrect answers included sulfur and silicon. In part 5 most students gave a correct equation with state symbols. In part 6, 26% of students gained both marks; the first mark was scored more often than the second. Common errors in the explanation included that the 2p electron lost from the sodium ion is higher in energy than the 3s electron in the magnesium ion and that full shells and/or orbitals are more stable than half-filled ones.

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